Lecture plan · for the lecturer

Backpropagation

AI · University · Lecture 2

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Draw a network as a graph.
- Backpropagate by hand.
- Explain vanishing gradients.

Activity: Three nodes, real numbers

1. Warm-up (5 min): Ask students to explain in one sentence how a network decides which weights to change.
2. Model (15 min): Work forward and backward through a three-node graph with numbers on the board.
3. Practise (20 min): Pairs backpropagate through one sigmoid neuron with squared loss.
4. Check (15 min): Compare each hand gradient with (L(w + h) - L(w - h)) / 2h in code.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Where did your hand gradient differ from the check?
- Why do residual connections help?
- What should students derive, and what can they import?

Exit ticket

- Backpropagate through $y = (2w)^2$ at $w = 1$.

Try it in your next lecture
Draw a three-node graph with real numbers; pairs compute every gradient by hand, then check against the framework's answer.

Seminar sheet · for students

Backpropagation

AI · University · Lecture 2

Name _____ Class _____ Date _____

1 **Why does the gradient shrink back through many sigmoid layers?**

- ◯ Each local derivative is ≤ 0.25 and they multiply
- ◯ The learning rate is too small
- ◯ Early layers have fewer weights

2 **Backpropagate through $L = (wx - y)^2$ with $w = 2$, $x = 3$, $y = 4$.**

The residual $wx - y$ = _____ , and $\partial L/\partial w = 2(wx - y)x$ = _____ .

Word bank: 2 · 6 · 4 · 12

3 **Which help gradients survive many layers? Choose all that apply.**

- ☐ ReLU activations
- ☐ Residual connections
- ☐ Careful initialisation
- ☐ Adding more sigmoid layers

4 **Computing all gradients by backpropagation costs roughly as much as one forward pass, not one pass per weight.**

- ◯ True
- ◯ False

5 **Match each student error to the misconception behind it.**

Adds local gradients instead of multiplying	• • • •	Confuses gradients with the optimiser
Forgets to sum gradients where a value is used twice	• • • •	Ignores fan-out in the graph
Thinks backprop updates the weights	• • • •	Misreads the chain rule

✎ **How do you currently teach backpropagation? Where could a concrete three-node example replace notation?**